

Technical Manual — ARV-5 Winch and Rigging Equipment

Reference TM-GEN-220 Revision B Unit: 12 Field Workshop

FICTIONAL CONTENT — generated for platform testing only.

1 Scope

This manual covers the main recovery winch, rope, and rigging accessories fitted to the fictional ARV-5 5-tonne recovery vehicle. It supplements the operator manual and does not replace first-line servicing schedules.

3.2 Winch Rope Specification

The main winch rope is 28 mm diameter, 6x36 construction with an independent wire rope core (IWRC). Nominal breaking load is 52,000 kgf. The rope shall be withdrawn from service and discarded when 10 or more broken wires are found within one lay length, or when any single strand is severed, whichever occurs first.

3.5 Rope Inspection Interval

Visually inspect the full working length of rope every 25 recovery operations, or every 90 days, whichever falls sooner. Record each inspection on form REC-14 and retain for 24 months.

5.1 Snatch Block

The issued snatch block has a safe working load of 16,000 kgf. Doubling the line through a snatch block halves the load on the rope and permits recovery of casualties up to the vehicle's rated capacity. The block shall not be used with rope showing any deformation of the outer strands.

5.4 Shackle and Strop Marking

All shackles shall carry a legible SWL stamp. Any shackle whose marking cannot be read shall be withdrawn. Recovery strops are colour-coded: blue for 8,000 kgf, green for 12,000 kgf, and red for 20,000 kgf.

7.2 Prohibited Practices

Do not use the recovery winch as a lifting appliance. Do not stand within 1.5 times the rope length of a rope under tension. Do not join two ropes with a bow shackle under load.